Jeongin Park

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Research Interest

Human-Computer Interaction, Augmented Reality, Human Centered AI

Education

Seoul National University *Sep 2025 – Present*
Seoul, South Korea

- M.S. in Computer Science and Engineering
- GPA: 4.3/4.3

Seoul National University *Mar 2021 – Aug 2025*
Seoul, South Korea

- B.S. in Computer Science and Engineering
- B.S. in Mathematical Sciences
- GPA: 4.0/4.3 (*Summa Cum Laude, Graduate Representative*)

Publications

DirectVis: Editing Code-Based Interactive Visualization with Direct Manipulation
Jeongin Park, Mingyu An, Hyunseo Yang, Junhyeong Hwangbo, Min Hyeong Kim, Hyeon Jeon, and Jinwook Seo
IEEE Pacific Visualization Conference 2026 (PacificVis'26), (Short Paper, 6 pages, Acceptance Rate: 35.6%)

Stop Misusing t-SNE and UMAP for Visual Analytics
Hyeon Jeon, Jeongin Park, Sungbok Shin, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics 2026 (VIS'26), (Full Paper)

Dataset-Adaptive Dimensionality Reduction
Hyeon Jeon, Jeongin Park, Soohyun Lee, Dae Hyun Kim, Sungbok Shin, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics 2025 (VIS'25), (Full Paper, 10 pages, Acceptance Rate: 24.4%)

Object-Oriented Prompting: A Framework for Precise and Scalable Image Generation
Jeongin Park, Hyeon Jeon, John Joon Young Chung, and Jinwook Seo
In Progress

Research Experience

Augmented Perception Lab, Carnegie Mellon University *Jan 2026 – Jul 2026*
Visiting Researcher - Advisor: Prof. David Lindlbauer *Pittsburgh, PA, USA*

- MorseQuery: Audio-only conversation support system. Designed a hearable interface for real-time semantic explanation and post-hoc summarization of missed speech.
- Investigating interaction techniques for target selection in speech-only interfaces.

Human Computer Interaction Lab, Seoul National University *Mar 2024 - Aug 2025*
Research Assistant - Advisor: Prof. Jinwook Seo *Seoul, South Korea*

- DirectVis: Designed and evaluated a direct manipulation-based interface for editing code-based visualizations; published at IEEE PacificVis 2026.
- Object-Oriented Prompting: Proposed a structured prompting framework leveraging object abstractions for controllable and scalable generation; manuscript in preparation.
- Led a USD 15,000 VR-based geometry education project, designing and implementing an interactive system and conducting a user study with four elementary school teachers to evaluate learning effectiveness.

Lunit Inc. *Dec 2023 - Feb 2024*
Internship – Advisors: Jaewoong Shin, Dr. Sergio Pereira *Seoul, South Korea*

- Developed *Attention-Based Cell Detection Enhancer*, improving traditional Whole Slide Image (WSI) cell classifier by integrating DETR, a vision transformer model, to capture spatial and feature-based relationships between cells.
- Optimized the cell interaction detection model by incorporating deformable attention layers to handle a larger number of cell predictions efficiently and using attention masking to construct precise and reliable cell interaction graph.

Max Planck Institute Computing and Data Facility *Sep 2023 - Dec 2023*
Internship – Advisors: Dr. Klaus Reuter, Dr. Markus Rampp *Garching, Germany*
(Fully Funded: airfare, accommodation, and insurance)

- Designed a parallelized data processing framework to analyze large-scale HPC data (50+ features, 50,000+ tasks/month), extracting time-series and statistical insights for resource optimization.
- Trained a Random Forest model for classifying job efficacy based on optimal use of computational resources and memory bandwidth, achieving a classification accuracy of 98.29%.
- Analyzed reasons for job inefficiency classification using explainable AI techniques including SHAP and Lime, confirming reliability by cross-checking with diverse methods and validating feature importance of the model.

Visual Computing Lab, Seoul National University *Mar 2023 - Aug 2023*
Undergraduate Research Opportunities Program – Advisor: Prof. Hanbyul Joo *Seoul, South Korea*

- Generated dense point clouds and meshes using Open3D, Colmap, and Meshlab after calibrating and rectifying stereo cameras to accurately compute intrinsic and extrinsic parameters (K, R, t) with MATLAB, ensuring precise 3D alignment.
- Applied the deep learning model HITNet for high-resolution, real-time depth estimation on images captured by stereo cameras, enabling accurate 3D reconstruction.

Scholarships and Awards

Seoul National University Leadership Award *Aug 2025*
- Awarded to 4 graduates among 2,500 in recognition of leadership contributions.

Presidential Science Scholarship *Mar 2024 - Aug 2025*
- Full Scholarship with an additional USD 2,000 stipend for academic encouragement.

KC Future Scholarship Foundation *Mar 2024 - Aug 2025*
- Full Scholarship with an additional USD 700 stipend for academic encouragement.

Semiconductor Specialized University Scholarship *Sep 2023 - Aug 2025*
- Awarded up to USD 8,500 for excellence in semiconductor studies.

National Science & Technology Scholarship *Mar 2023 - Mar 2024*
- Full Scholarship covering tuition for outstanding students.

Academic Excellence Scholarship, Seoul National University *Mar 2021 - Feb 2023*
- Received four times in recognition of academic excellence from 2021 to 2022.

Projects

SNU Social Responsibility "For All" Project *Mar 2025 – Jun 2025*
- Developed a mobile application to support independent museum and exhibition visits for individuals with visual impairments.
- Designed and implemented a computer vision-based indoor localization system and a user-centered navigation interface.

BugRider: 3D Adventure Game Development *Mar 2025 – Jun 2025*
- Led a five-member development team, overseeing project planning, task allocation, and development schedule management.
- Created 3D assets and animations using Blender, and developed game controls and audio systems in Unity.

VR-Based Geometry Education Tool *Dec 2024 - April 2025*
- Secured a research grant of USD 15,000 by independently writing and submitting the full project proposal.
- Designed and developed an immersive VR educational tool aimed at improving geometry learning outcomes.
- Conducted a user study with four elementary school teachers to evaluate usability, effectiveness, and pedagogical integration.
- Responsibilities included project planning, content design, system implementation, and experimental evaluation.

Question-Answer-and-Connect [GitHub] [Website] *Mar 2024 - June 2024*
- Developed a real-time framework for large-scale educational group chats to reduce information overload, categorizing messages into questions and answers while linking related content for better accessibility.
- Leveraged OpenAI's API for dynamic response generation, enhancing accessibility and usability in group communication.

Ko-Radiology-GPT [GitHub] [Website] *Mar 2024 - Jun 2024*
- Fine-tuned a Korean GPT model for radiology report interpretation using curated MIMIC-CXR and AIHub datasets.
- Improved performance by 28% over untrained Llama2 and 130% over English-tuned models for radiology-specific tasks.

MAPSEE – A map-based journal app [GitHub] [Website] *Sep 2022 - Aug 2023*
- Developed Mapsee, a map-based journal application enabling users to create and share location-tagged diaries.

- Implemented efficient map-based search, optimized caching, and integrated storage APIs using React Native, Google Maps API, and Firebase.
- Mentored junior developers, teaching React Native, Git workflows, and Google API integration.

Additional Experiences

SNU Tomorrow's Engineers Membership(STEM; Engineering Honor Society), Member *Mar 2023 - Aug 2025*

- Hosted engineering seminars and networking events, including alumni meetups and collaborations with external organizations, to promote knowledge exchange and connections.

Northeast Asia Student Round Table (SRT), Chairman *Sep 2023 - Aug 2024*

- Organized 30+ initiatives promoting dialogue among students from four countries, leading discussions on educational disparities and sharing Korean traditions like cuisine and games to foster cultural exchange.

SNU Student Ambassadors (SSA), Vice President *Mar 2022 - Dec 2022*

- Coordinated events for foreign dignitaries, managed school-student communications, and oversaw accounting for ambassador activities.

Language Proficiency

TOEFL: 6.0 (118/120; Reading 6.0, Listening 5.5, Speaking 6.0, Writing 5.5).